

Major Code and Data Migrations for GRNsight, a Web Application for Visualizing Gene Regulatory and Protein-Protein Interaction Network Models

Cecilia Zaragoza & Ngoc Kim Ngan Tran
Department of Computer Science
Loyola Marymount University

LMU Undergraduate Symposium
March 20, 2026



Loyola Marymount University
Frank R. Seaver
College of Science
and Engineering

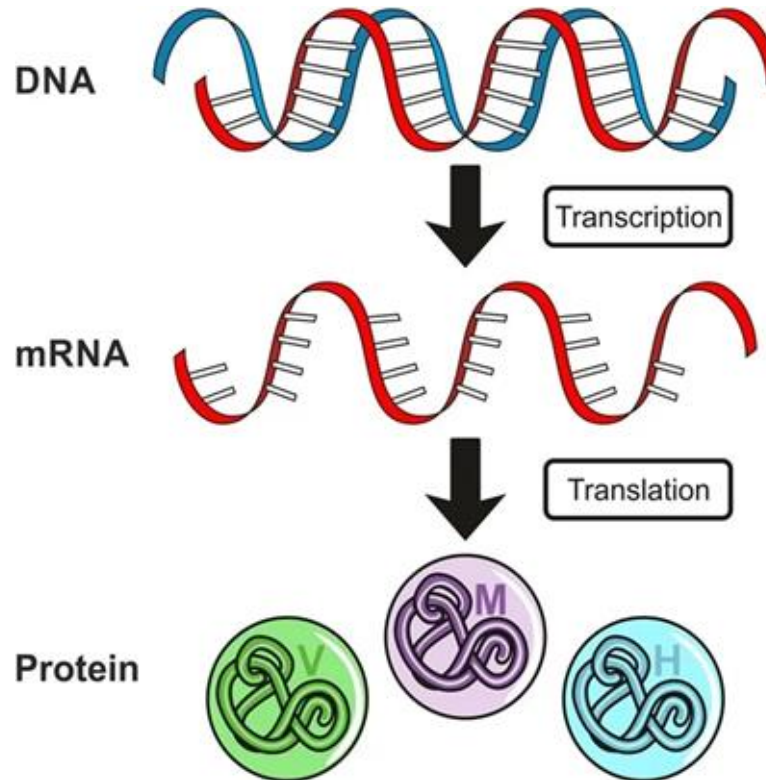
Outline

- GRNsight is an open-source tool that allows for easier visualization of Gene Regulatory Network (GRN) and Protein Protein Interactions (PPI).
- GRNsight has migrated from JavaScript frontend to React Vite and Grommet UI.
- We transitioned our primary data source from YeastMine to AllianceMine.
- We re-engineered database pipeline into a modular architectural pattern.
- The primary goal of this modernization is to provide a faster, more scalable, and intuitive platform for biological network visualization.

Outline

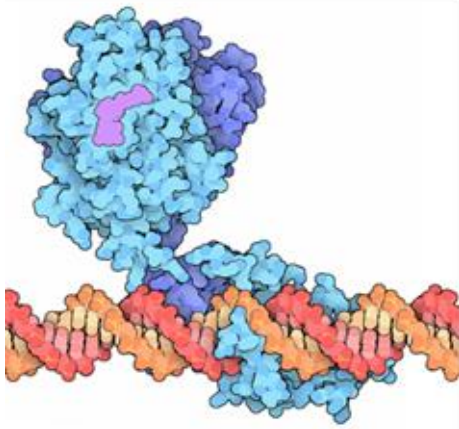
- GRNsight is an open-source tool that allows for easier visualization of Gene Regulatory Network (GRN) and Protein Protein Interactions (PPI).
- GRNsight has migrated from JavaScript frontend to React Vite and Grommet UI.
- We transitioned our primary data source from YeastMine to AllianceMine.
- We re-engineered database pipeline into a modular architectural pattern.
- The primary goal of this modernization is to provide a faster, more scalable, and intuitive platform for biological network visualization.

Gene expression is the flow of information in a cell from DNA to RNA to protein

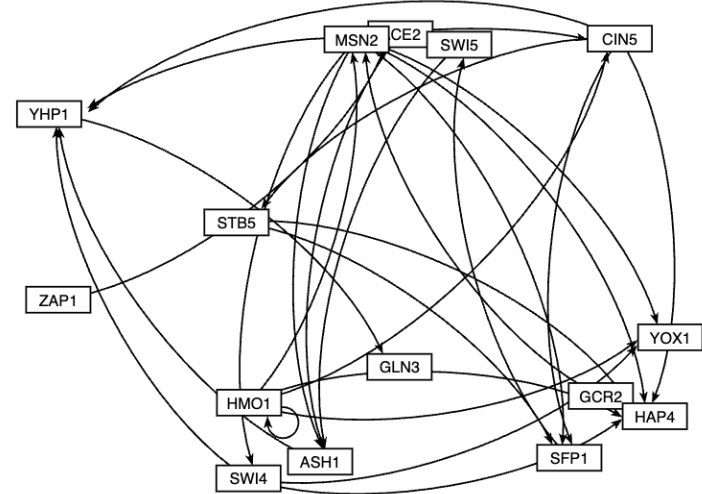


A Gene Regulatory Network (GRN) consists of genes, transcription factors and the regulatory connections between them which govern the level of expression of mRNA and protein from genes

Regulatory Transcription Factors bind to DNA and turn on (activate) or turn off (repress) gene expression

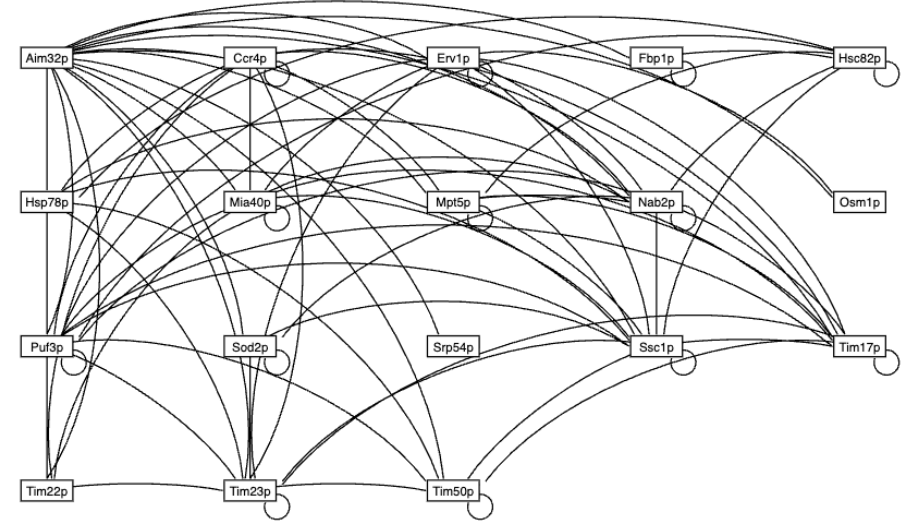
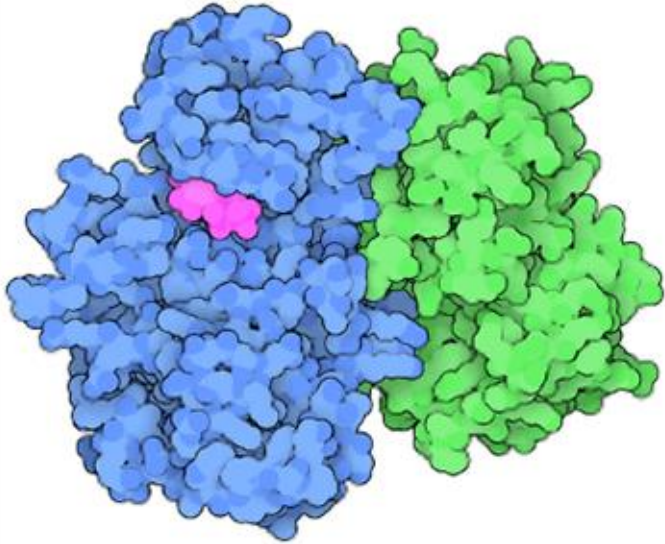


“PDB101: Molecule of the Month: Vitamin D receptor,”
RCSB: PDB-101. <https://pdb101.rcsb.org/motm/155>



- Node = transcription factor / gene
- Arrow edge = regulatory relationship

Protein-Protein Interactions refer to physical interactions of proteins binding to each other - the basis of nearly all cell functions



"PDB101: Molecule of the Month: Cyclin and cyclin-dependent kinase,"
RCSB: PDB-101. <https://pdb101.rcsb.org/motm/236>

- Node = proteins
- Edge = binding relationship

GRNsight is an integral part of a data analysis workflow for biologists who model GRNs



GRNsight

Web app for visualizing models of small-scale gene regulatory and protein-protein interaction networks.

A Joint Project of the
LMU Bioinformatics and
Biomathematics Groups

LMU LA
Loyola Marymount
University

Home
About
News
Developers
Documentation
Wiki
Publications
People
Contact
Links
Beta
Privacy

Last modified:
2/17/2026

GRNsight v7.4.1

Welcome to GRNsight. Please select a network to view.

Network Layout Node Edge View Export Help Demo Demo #1: Unweighted GRN (15 genes, 28 edges)

15 nodes
28 edges

Network

Network Source

Demo

Demo #1: Unweighted

Open File

Load from Database

Reload

Network Mode:

Gene Regulatory Network

Species:

Saccharomyces cerevisiae

Layout

Force Graph Grid Layout

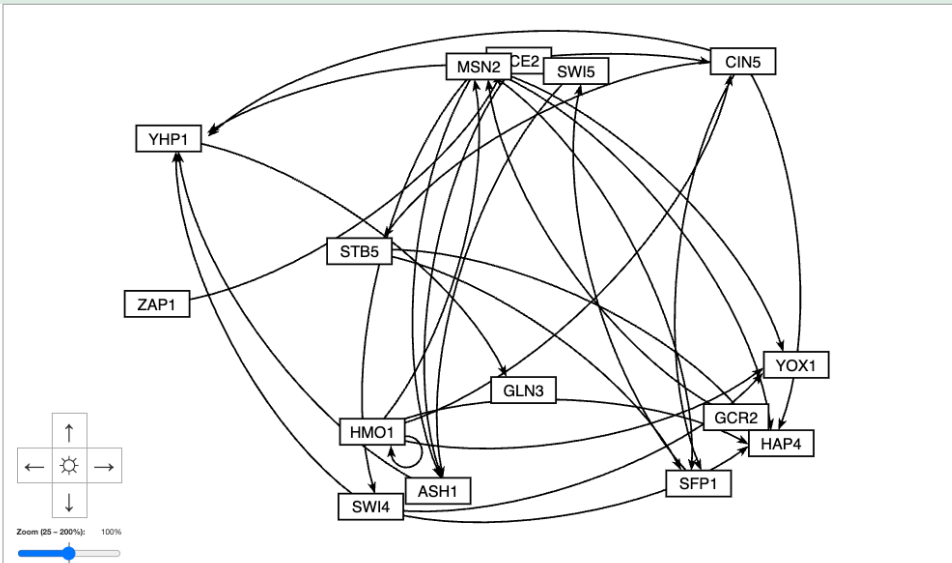
Link Distance (1-1000): 500

Charge (-2000-0): -50

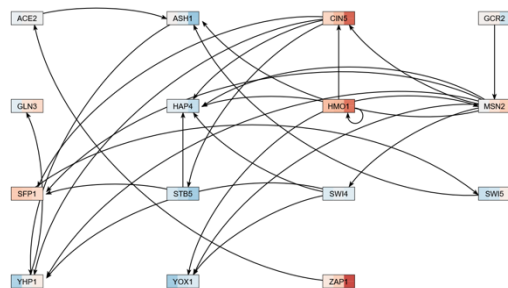
Lock Force Parameters

Reset Force Parameters

Undo Reset



GRNsight is an integral part of a data analysis workflow for biologists who model GRNs



Select network and
expression data from
GRNsight database

Load from Database

Generate Network

Warning: changing network type or source will clear the list of selected genes or proteins below.

Network Type:

Network Source:

Select gene:

Add new genes go below! Click on a gene to remove it.

ACE2 (YAL00112C)	ASH1 (YAL00400C)	CINS (YAL00402C)
GCR2 (YAL00402C)	GLND (YAL00402C)	HAP4 (YAL00402C)
HMO1 (YAL00402C)	MGN2 (YAL00402C)	SFP1 (YAL00402C)
STB5 (YAL00402C)	SWH (YAL00402C)	SWI5 (YAL00402C)
YHP1 (YAL00402C)	YOX1 (YAL00402C)	

Generate Network Cancel

Node

☒ Enable Node Coloring

Select from user-uploaded
expression data, or use data
from our Expression Database

Top Dataset

☒ Average Replicate Values

Bottom Dataset

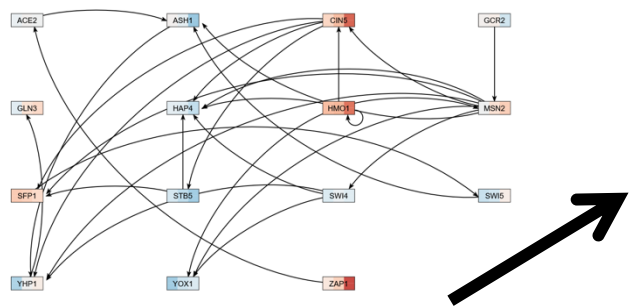
☒ Average Replicate Values

Log Fold Change Max Value
(0.01-100):

Set

-3.50 0 3.50

GRNsight is an integral part of a data analysis workflow for biologists who model GRNs



Export to Excel



A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	id	production_rate													
2	id	A	B	C											
3	id	A	B	C											
4	id	A	B	C											
5	id	A	B	C											
6	id	A	B	C											
7	id	A	B	C											
8	id	A	B	C											
9	id	A	B	C											
10	id	A	B	C											
11	id	A	B	C											
12	id	A	B	C											
13	id	A	B	C											
14	id	A	B	C											
15	id	A	B	C											
16	id	A	B	C											
17	id	A	B	C											
18	id	A	B	C											
19	id	A	B	C											
20	id	A	B	C											
21	id	A	B	C											
22	id	A	B	C											
23	id	A	B	C											
24	id	A	B	C											
25	id	A	B	C											
26	id	A	B	C											
27	id	A	B	C											
28	id	A	B	C											
29	id	A	B	C											
30	id	A	B	C											
31	id	A	B	C											
32	id	A	B	C											
33	id	A	B	C											
34	id	A	B	C											
35	id	A	B	C											
36	id	A	B	C											
37	id	A	B	C											
38	id	A	B	C											
39	id	A	B	C											
40	id	A	B	C											
41	id	A	B	C											
42	id	A	B	C											
43	id	A	B	C											
44	id	A	B	C											
45	id	A	B	C											
46	id	A	B	C											
47	id	A	B	C											
48	id	A	B	C											
49	id	A	B	C											
50	id	A	B	C											
51	id	A	B	C											
52	id	A	B	C											
53	id	A	B	C											
54	id	A	B	C											
55	id	A	B	C											
56	id	A	B	C											
57	id	A	B	C											
58	id	A	B	C											
59	id	A	B	C											
60	id	A	B	C											
61	id	A	B	C											
62	id	A	B	C											
63	id	A	B	C											
64	id	A	B	C											
65	id	A	B	C											
66	id	A	B	C											
67	id	A	B	C											
68	id	A	B	C											
69	id	A	B	C											
70	id	A	B	C											
71	id	A	B	C											
72	id	A	B	C											
73	id	A	B	C											
74	id	A	B	C											
75	id	A	B	C											
76	id	A	B	C											
77	id	A	B	C											
78	id	A	B	C											
79	id	A	B	C											
80	id	A	B	C											
81	id	A	B	C											
82	id	A	B	C											
83	id	A	B	C											
84	id	A	B	C											
85	id	A	B	C											
86	id	A	B	C											
87	id	A	B	C											
88	id	A	B	C											
89	id	A	B	C											
90	id	A	B	C											
91	id	A	B	C											
92	id	A	B	C											
93	id	A	B	C											
94	id	A	B	C											
95	id	A	B	C											
96	id	A	B	C											
97	id	A	B	C											
98	id	A	B	C											
99	id	A	B	C											
100	id	A	B	C											

Export to Excel

Select the Expression Data Source:

☐ None

☐ Agilent_2012

☐ Burren_2012

☐ Dahlquist_2018

☐ Klingman_2002

☐ Thomas_2007

Select Workbook Sheets to Export:

☒ Select All

Network Sheets:

☒ network

☐ network_optimized_weights

Expression Sheets:

☒ dahlquist_2018_wt

☒ dahlquist_2018_wt_optimized_weights

☒ dahlquist_2018_wt_optimized_weights

☒ dahlquist_2018_wt_optimized_weights

☒ wt_2018_wt

Additional Sheets:

☒ degradation_rates

☒ optimization_parameters

☒ production_rates

☒ threshold_b

Export Workbook Cancel

Select network and expression data from GRNsight database

Load from Database

Generate Network

Warning: changing network type or source will clear the list of selected genes or proteins below.

Network Type:

Network Source:

Select gene:

Add new genes go below! Click on a gene to remove it.

ACE2 (YJL012C)	ASH1 (YJL012C)	CINS (YJL012C)
GCR2 (YJL012C)	GLN3 (YJL012C)	HAP4 (YJL012C)
HMO1 (YJL012C)	MSH2 (YJL012C)	SFP1 (YJL012C)
STB5 (YJL012C)	SWH4 (YJL012C)	SWH5 (YJL012C)
YOP1 (YJL012C)	YOK1 (YJL012C)	ZAP1 (YJL012C)

Generate Network Cancel

Node

☒ Enable Node Coloring

Select from user-uploaded expression data, or use data from our Expression Database

Top Dataset

☒ Average Replicate Values

Bottom Dataset

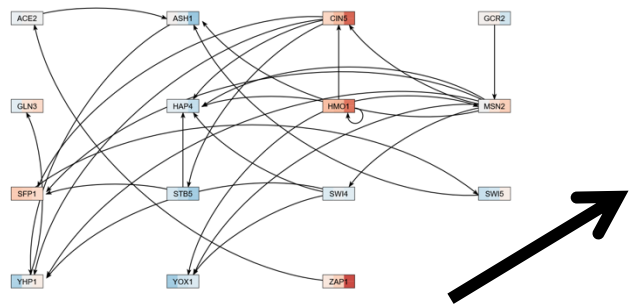
☒ Average Replicate Values

Log Fold Change Max Value
(0.01-100):

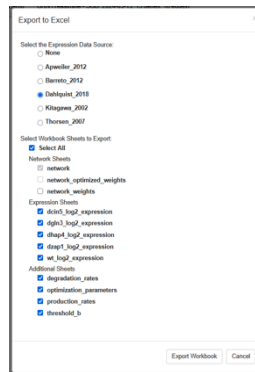
Set

-3.50 0 3.50

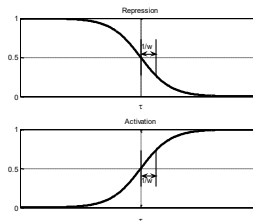
GRNsight is an integral part of a data analysis workflow for biologists who model GRNs



Export to Excel



Dynamical systems modeling using GRNmap



Select network and expression data from GRNsight database

Load from Database

Generate Network

Warning: changing network type or source will clear the list of selected genes or proteins below.

Network Type:

Network Source:

Select gene:

Add new genes go below! Click on a gene to remove it.

ACE2 (YJL017C)	ASH1 (YJL048C)	CRN5 (YJL049C)
GCR2 (YJL018C)	GLN3 (YJL019C)	HAP4 (YJL020C)
HMO1 (YJL021C)	MNR2 (YJL022C)	SFP1 (YJL023C)
STB5 (YJL024C)	SWH4 (YJL025C)	SWH5 (YJL026C)
YHP1 (YJL027C)	YOK1 (YJL028C)	ZAP1 (YJL029C)

Generate Network Cancel

Node

☒ Enable Node Coloring

Select from user-uploaded expression data, or use data from our Expression Database

Top Dataset

☒ Average Replicate Values

Bottom Dataset

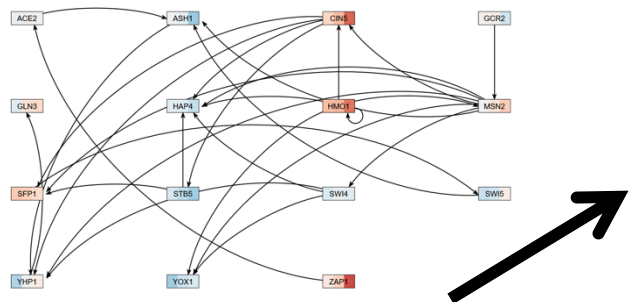
☒ Average Replicate Values

Log Fold Change Max Value
(0.01-100):

Set

-3.50 0 3.50

GRNsight is an integral part of a data analysis workflow for biologists who model GRNs



Export to Excel



1	A B C			B C D E F G H I J K L M N													
2	id	production_rate															
3	id	degradation_rate															
4	A B B C			B C D E F G H I J K L M N													
5	id	threshold_b															
6	id	threshold_b															
7	id	threshold_b															
8	id	threshold_b															
9	id	threshold_b															
10	id	threshold_b															
11	id	threshold_b															
12	id	threshold_b															
13	id	threshold_b															
14	id	threshold_b															
15	id	threshold_b															
16	id	threshold_b															
17	id	threshold_b															
18	id	threshold_b															
19	id	threshold_b															
20	id	threshold_b															
21	id	threshold_b															
22	id	threshold_b															
23	id	threshold_b															
24	id	threshold_b															
25	id	threshold_b															
26	id	threshold_b															
27	id	threshold_b															
28	id	threshold_b															
29	id	threshold_b															
30	id	threshold_b															
31	id	threshold_b															
32	id	threshold_b															
33	id	threshold_b															
34	id	threshold_b															
35	id	threshold_b															
36	id	threshold_b															
37	id	threshold_b															
38	id	threshold_b															
39	id	threshold_b															
40	id	threshold_b															
41	id	threshold_b															
42	id	threshold_b															
43	id	threshold_b															
44	id	threshold_b															
45	id	threshold_b															
46	id	threshold_b															
47	id	threshold_b															
48	id	threshold_b															
49	id	threshold_b															
50	id	threshold_b															
51	id	threshold_b															
52	id	threshold_b															
53	id	threshold_b															
54	id	threshold_b															
55	id	threshold_b															
56	id	threshold_b															
57	id	threshold_b															
58	id	threshold_b															
59	id	threshold_b															
60	id	threshold_b															
61	id	threshold_b															
62	id	threshold_b															
63	id	threshold_b															
64	id	threshold_b															
65	id	threshold_b															
66	id	threshold_b															
67	id	threshold_b															
68	id	threshold_b															
69	id	threshold_b															
70	id	threshold_b															
71	id	threshold_b															
72	id	threshold_b															
73	id	threshold_b															
74	id	threshold_b															
75	id	threshold_b															
76	id	threshold_b															
77	id	threshold_b															
78	id	threshold_b															
79	id	threshold_b															
80	id	threshold_b															
81	id	threshold_b															
82	id	threshold_b															
83	id	threshold_b															
84	id	threshold_b															
85	id	threshold_b															
86	id	threshold_b															
87	id	threshold_b															
88	id	threshold_b															
89	id	threshold_b															
90	id	threshold_b															
91	id	threshold_b															
92	id	threshold_b															
93	id	threshold_b															
94	id	threshold_b															
95	id	threshold_b															
96	id	threshold_b															
97	id	threshold_b															
98	id	threshold_b															
99	id	threshold_b															
100	id	threshold_b															

Export to Excel

Select the Expression Data Source:

- ☐ None
- ☐ Agreuter_2012
- ☐ Baranyi_2012
- ☒ Dahlquist_2018
- ☐ Klingmann_2002
- ☐ Thomas_2007

Select Workbook Sheets to Export:

Select All

Network Sheets:

- ☒ network
- ☐ network_optimized_weights

Expression Sheets:

- ☒ dahlquist_2018_wt
- ☒ dahlquist_2018_wt
- ☒ dahlquist_2018_wt
- ☒ dahlquist_2018_wt
- ☒ dahlquist_2018_wt

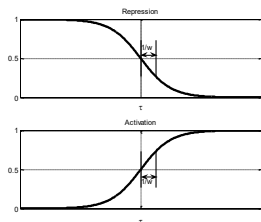
Additional Sheets:

- ☒ degradation_rates
- ☒ optimization_parameters
- ☒ production_rates
- ☒ threshold_b

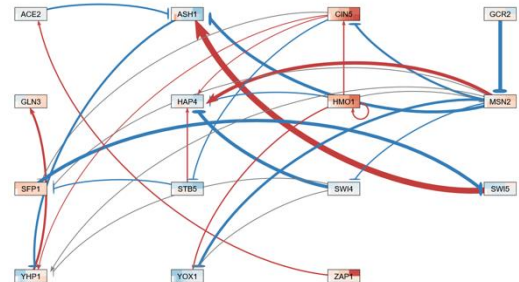
Export Workbook Cancel

Select network and expression data from GRNsight database

Dynamical systems modeling using GRNmap



Visualization of modeling results using GRNsight



Load from Database

Generate Network

Warning: changing network type or source will clear the list of selected genes or proteins below.

Network Type:

Network Source:

Select gene:

Add new genes go below! Click on a gene to remove it.

ACE2 (YAL0012C)	ASH1 (YAL0080C)	CIN5 (YAL0082C)
GCR2 (YAL0084C)	GLN3 (YAL0086C)	HAP4 (YAL0088C)
HMO1 (YAL0090C)	MSN2 (YAL0092C)	SFP1 (YAL0094C)
STB5 (YAL0096C)	SWH4 (YAL0098C)	SWI5 (YAL0100C)
YHP1 (YAL0102C)	YOX1 (YAL0104C)	ZAP1 (YAL0106C)

Generate Network Cancel

Node

☒ Enable Node Coloring

Select from user-uploaded expression data, or use data from our Expression Database

Top Dataset:

☒ Average Replicate Values

Bottom Dataset:

☒ Average Replicate Values

Log Fold Change Max Value (0.01-100):

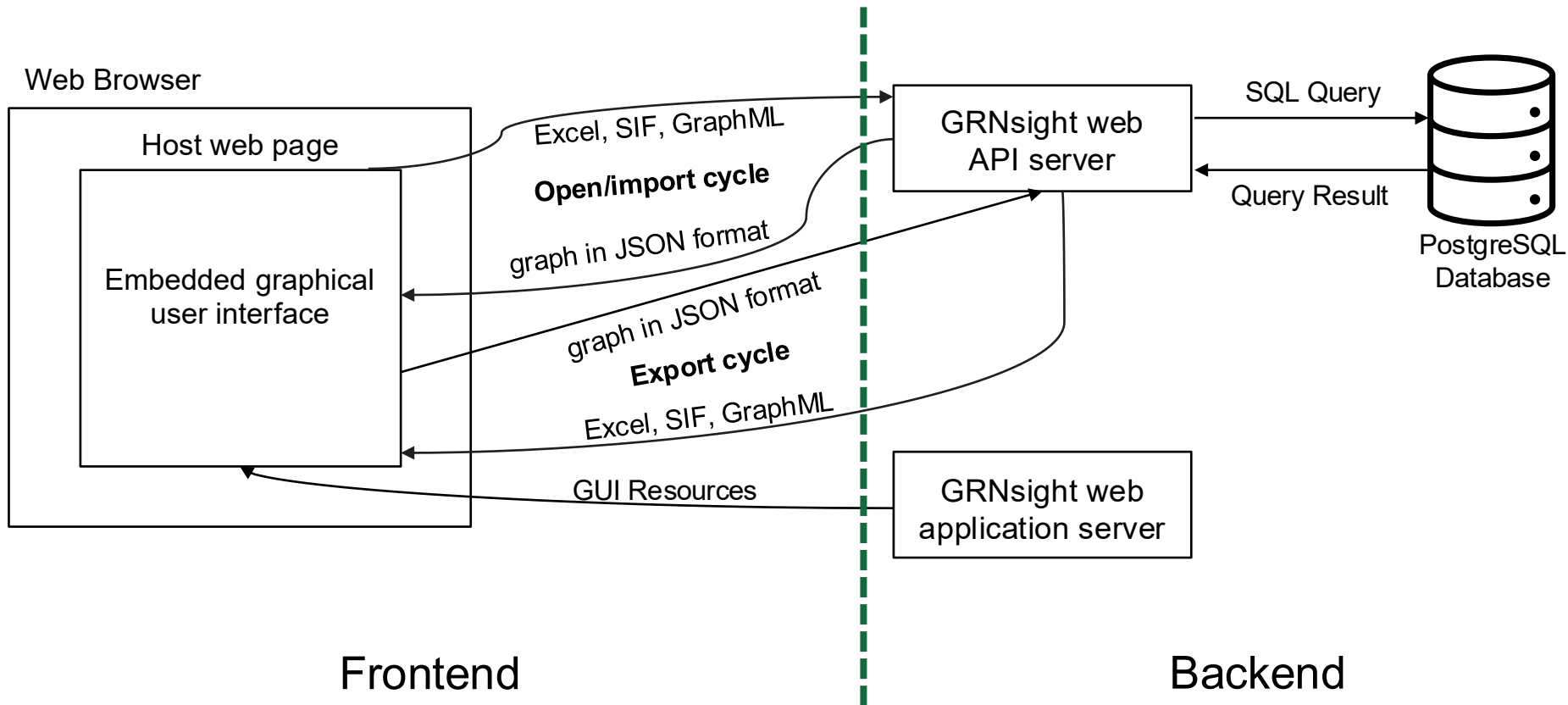
Set

-3.50 0 3.50

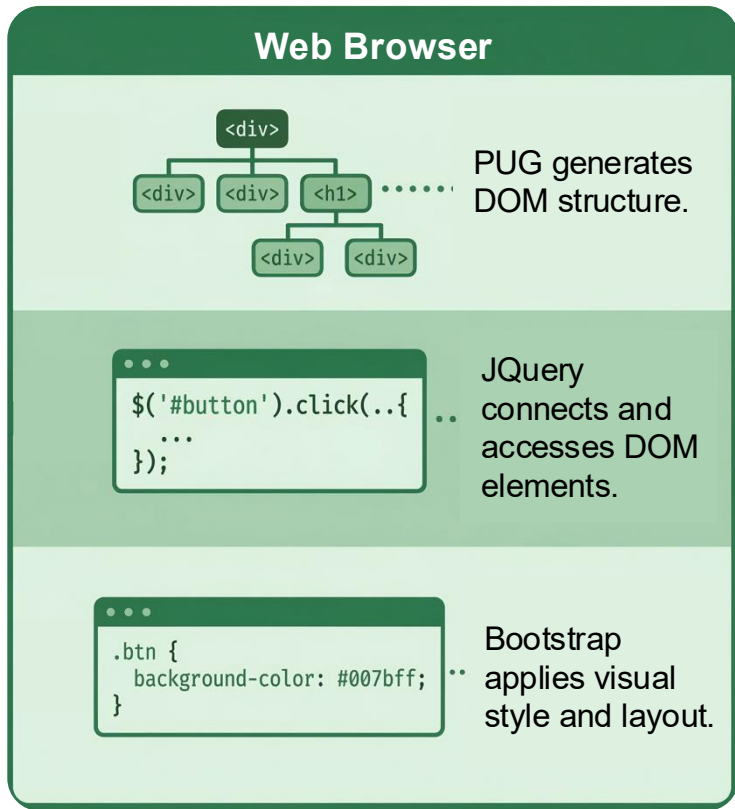
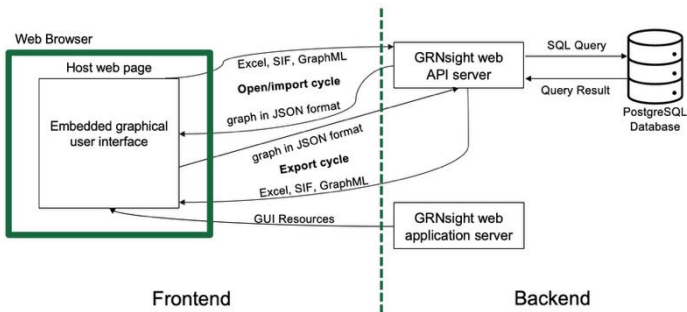
Outline

- GRNsight is an open-source tool that allows for easier visualization of Gene Regulatory Network (GRN) and Protein Protein Interactions (PPI).
- GRNsight has migrated from JavaScript frontend to React Vite and Grommet UI.
- We transitioned our primary data source from YeastMine to AllianceMine.
- We re-engineered database pipeline into a modular architectural pattern.
- The primary goal of this modernization is to provide a faster, more scalable, and intuitive platform for biological network visualization.

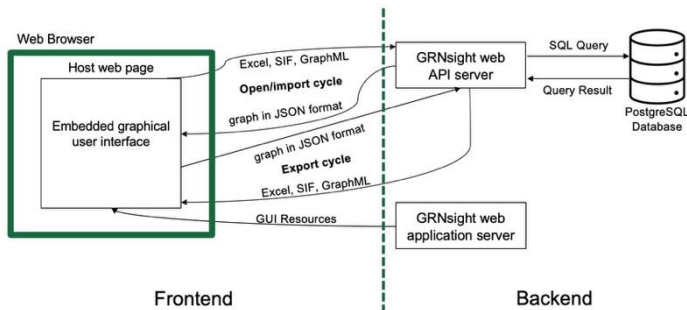
GRNsight serves the browser UI from the web application server while the API server handles import/export and all database access.



GRNsight Classic built on jQuery/Bootstrap/Pug framework for frontend resulting in code redundancy and complex UI updates



React/Grommet framework for frontend improves state management, code maintainability, and user interaction



Web Browser

```
<Button primary onClick={() => {}}
  icon=<Add />
  label="ADD ITEM"
/>
```

React JSX allows creation, access, and manipulation of DOM elements in the same place.

```
<Button
  primary
  onClick={() => {}}
  label="Styled button"
/>
```

Card

Styled button

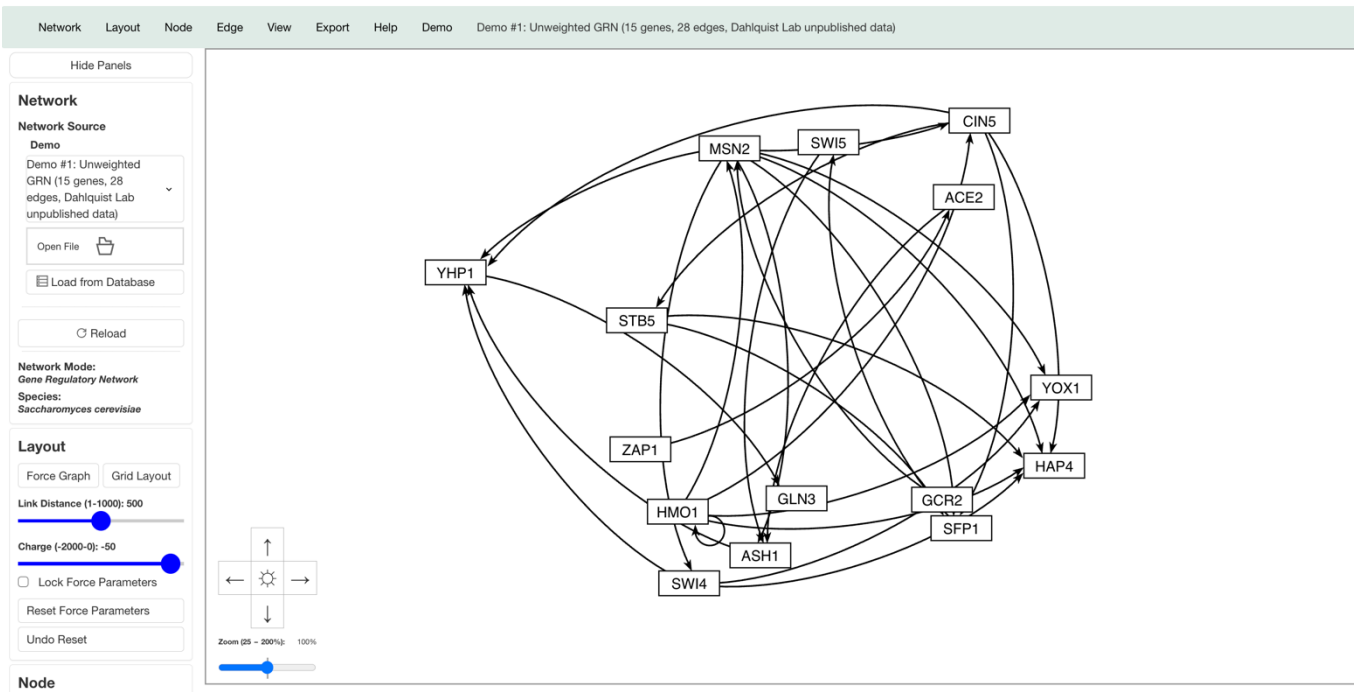
Grommet provides styling and theming via reusable components.



GRNsight React went live on October 15, 2025, and can now display graphs

GRNsight React: v0.0.3 (03/04/2026)

Disclaimer: This version of GRNsight is currently under development and is unstable. For the most stable version, please go to the [GRNsight home page](#).



- <https://dondi.github.io/GRNsight/react-thesis-4081/>
- Browser: Google Chrome

React allows the fast implementation of collapsible side panels

GRNsight v7.4.1

Welcome to GRNsight. Please select a network to view.

NetworkLayoutNodeEdgeViewExportHelpDemo

Network

Network Source

Demo

Demo #1: Unweighted C

Open File

Load from Database

Reload

Network Mode:

Gene Regulatory Network

Species:

Saccharomyces cerevisiae

Layout

Force GraphGrid Layout

Link Distance (1-1000): 500

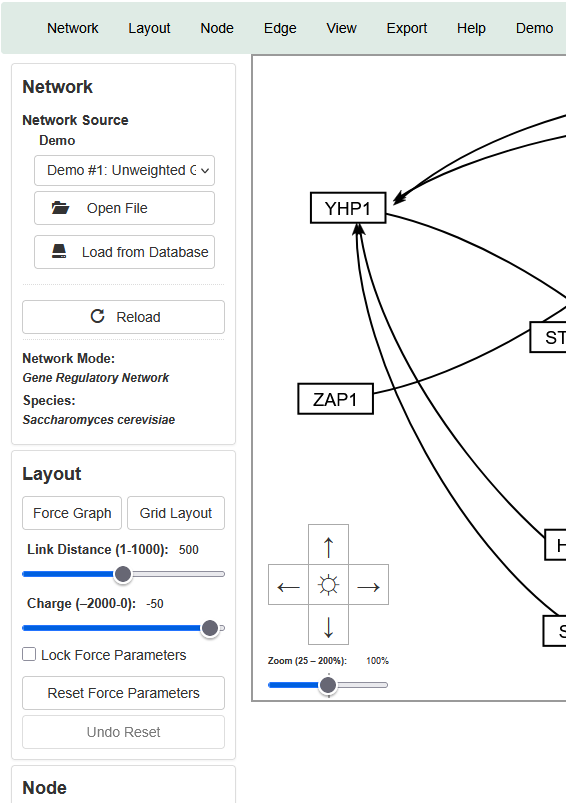
Charge (-2000-0): -50

Lock Force Parameters

Reset Force Parameters

Undo Reset

Node



GRNsight React: v0.0.4 (03/12/2026)

Disclaimer: This version of GRNsight is currently under de

NetworkLayoutNodeEdgeView

Hide Panels

Network

Network Source

Demo

Demo #1: Unweighted GRN (15 genes, 28 edges, Dahlquist Lab unpublished data)

Open File

Load from Database

Reload

Network Mode:

Gene Regulatory Network

Species:

Saccharomyces cerevisiae

Layout

Force GraphGrid Layout

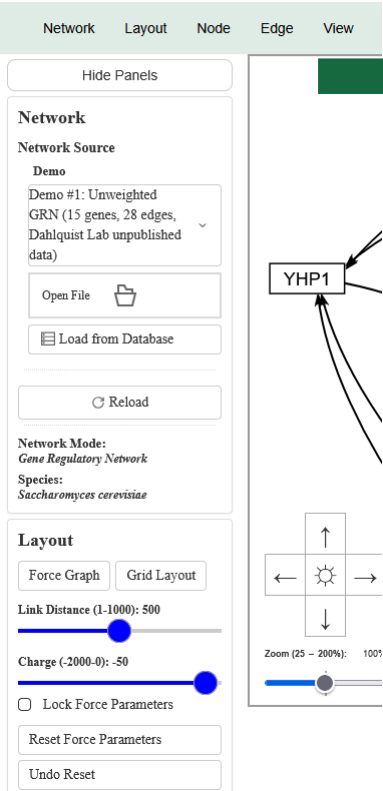
Link Distance (1-1000): 500

Charge (-2000-0): -50

Lock Force Parameters

Reset Force Parameters

Undo Reset



GRNsight React: v0.0.4 (03/12/2026)

Disclaimer: This version of GRNsight is currently under development and is unstable. For

NetworkLayoutNodeEdgeViewExportHelpDemo

Network

Network Source

Demo

Demo #1: Unweighted GRN (15 genes, 28 edges, Dahlquist Lab unpublished data)

Open File

Load from Database

Reload

Network Mode:

Gene Regulatory Network

Species:

Saccharomyces cerevisiae

Layout

Force GraphGrid Layout

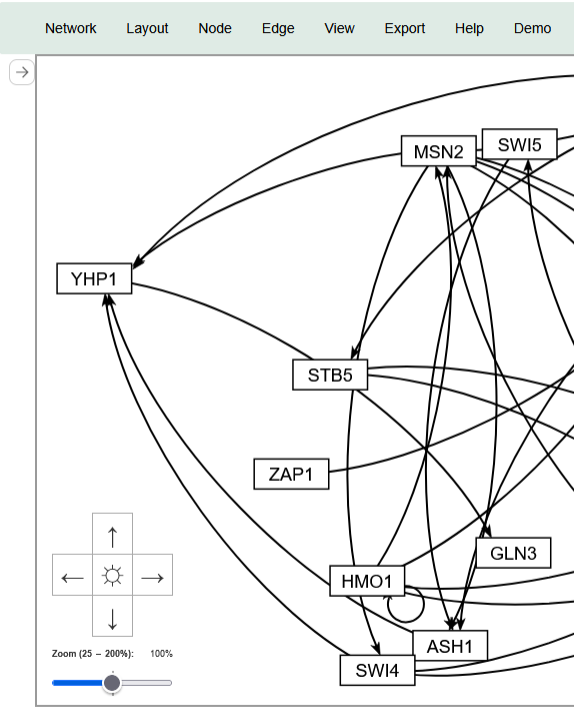
Link Distance (1-1000): 500

Charge (-2000-0): -50

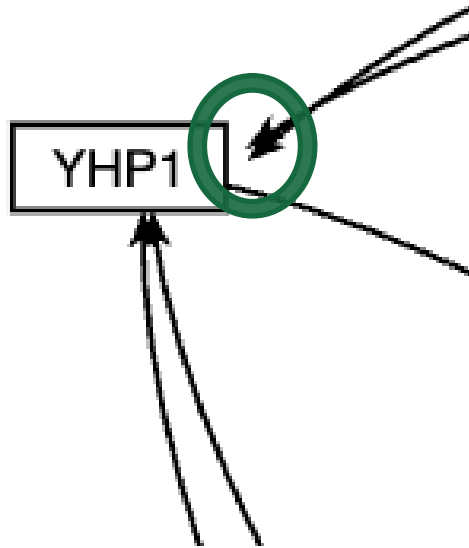
Lock Force Parameters

Reset Force Parameters

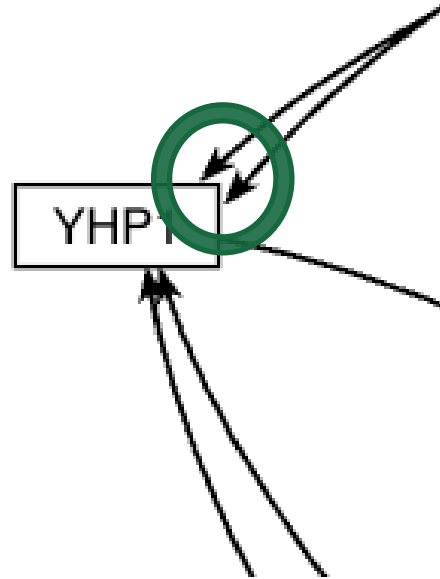
Undo Reset



The high-level code review facilitated the fixing of a long-standing arrowhead bug in GRNsight Classic

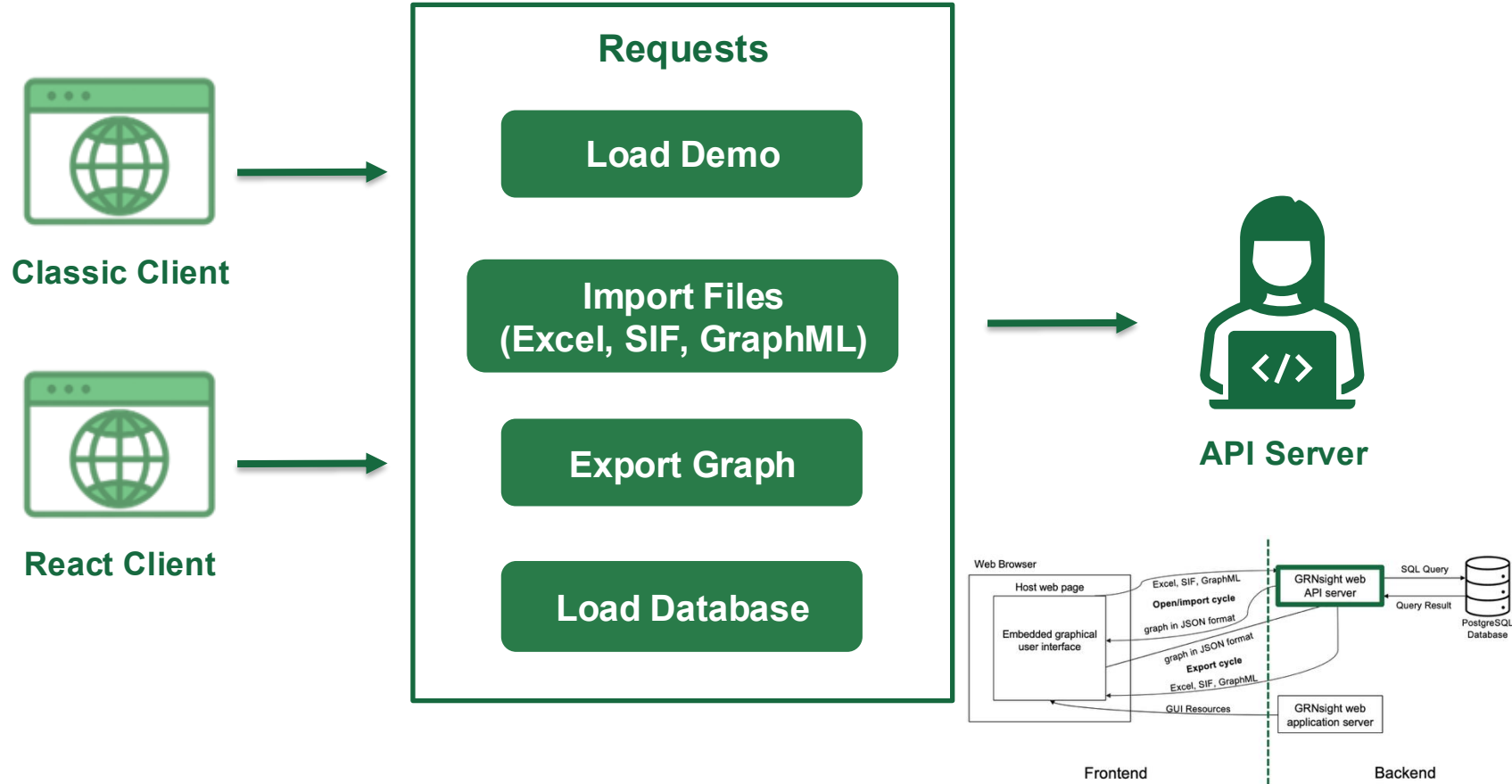


Classic Arrowheads with space between the perimeter

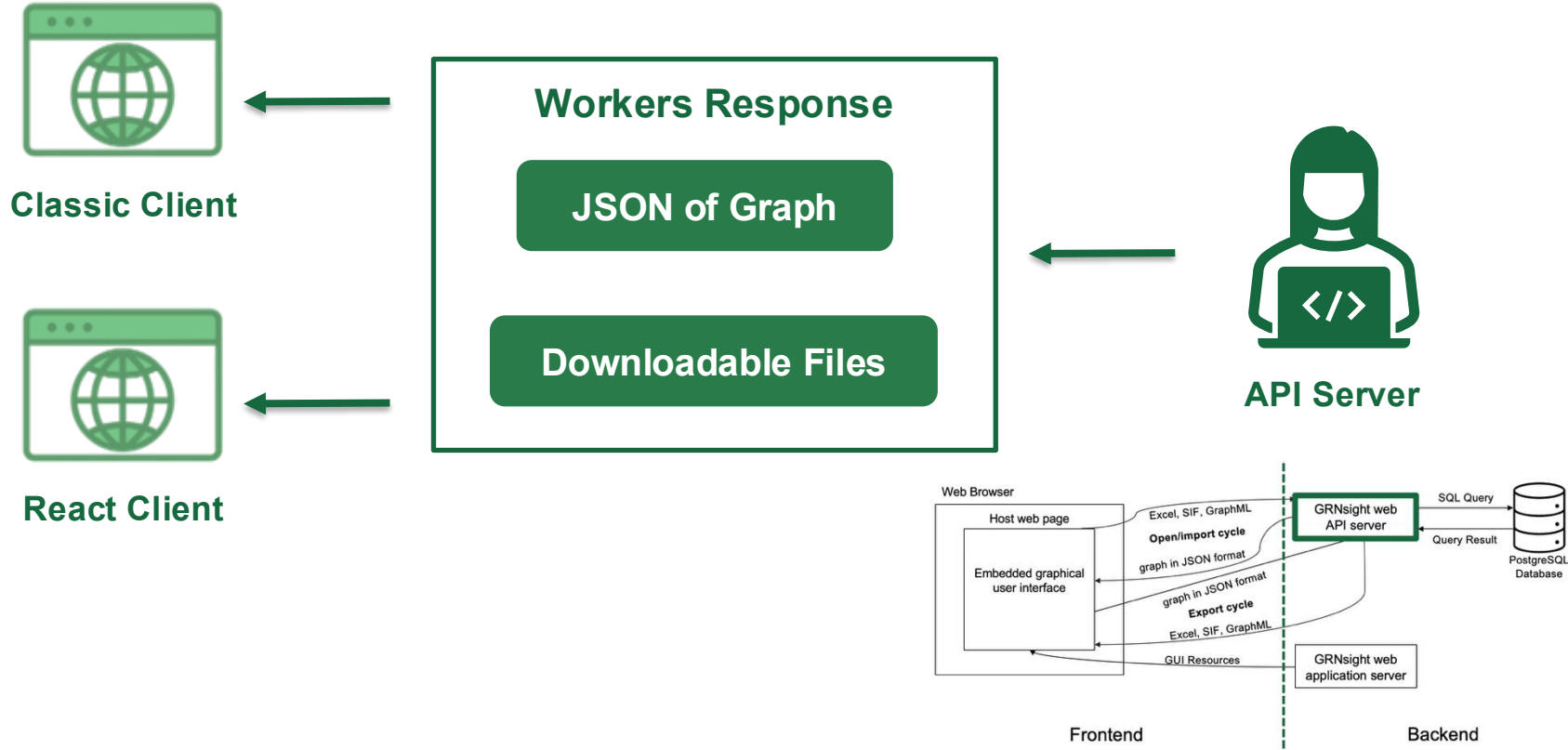


React Arrowheads touching the perimeter of the node

Clients send different requests to the workers in the back



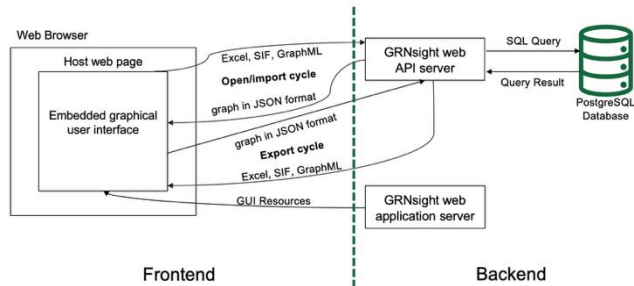
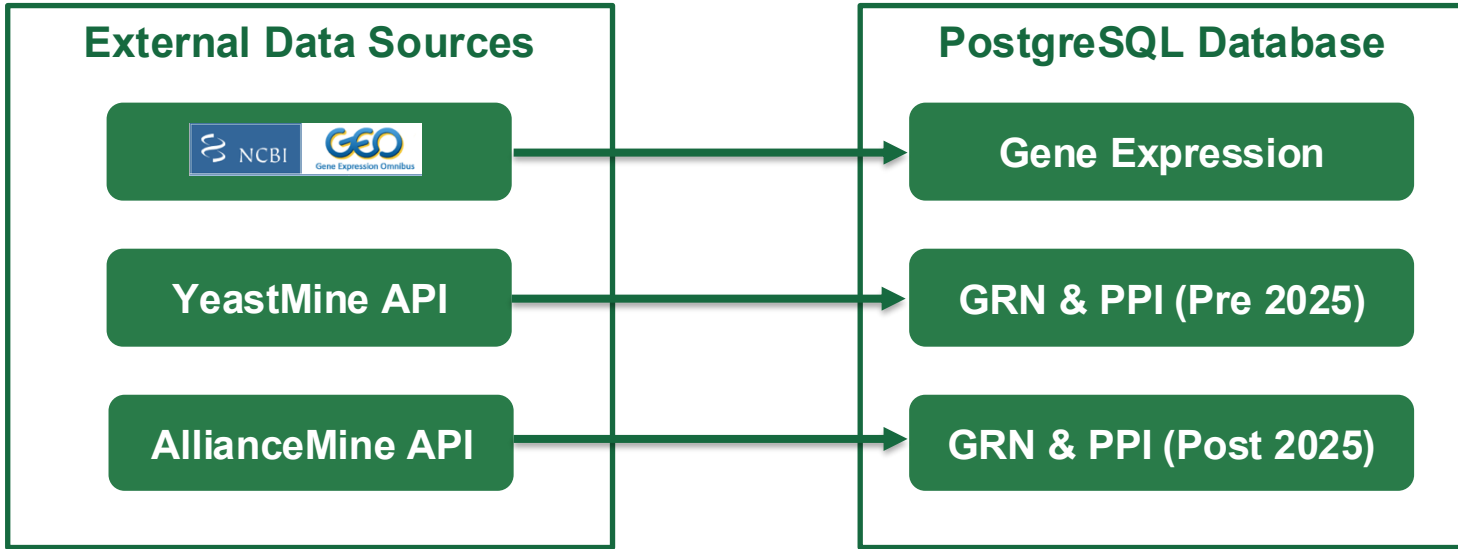
Workers process the requests and return proper results to clients



Outline

- GRNsight is an open-source tool that allows for easier visualization of Gene Regulatory Network (GRN) and Protein Protein Interactions (PPI).
- GRNsight has migrated from JavaScript frontend to React Vite and Grommet UI.
- We transitioned our primary data source from YeastMine to AllianceMine.
- We re-engineered database pipeline into a modular architectural pattern.
- The primary goal of this modernization is to provide a faster, more scalable, and intuitive platform for biological network visualization.

GRNsight's database currently uses 3 different sources for gene expression data, GRN and PPI network data



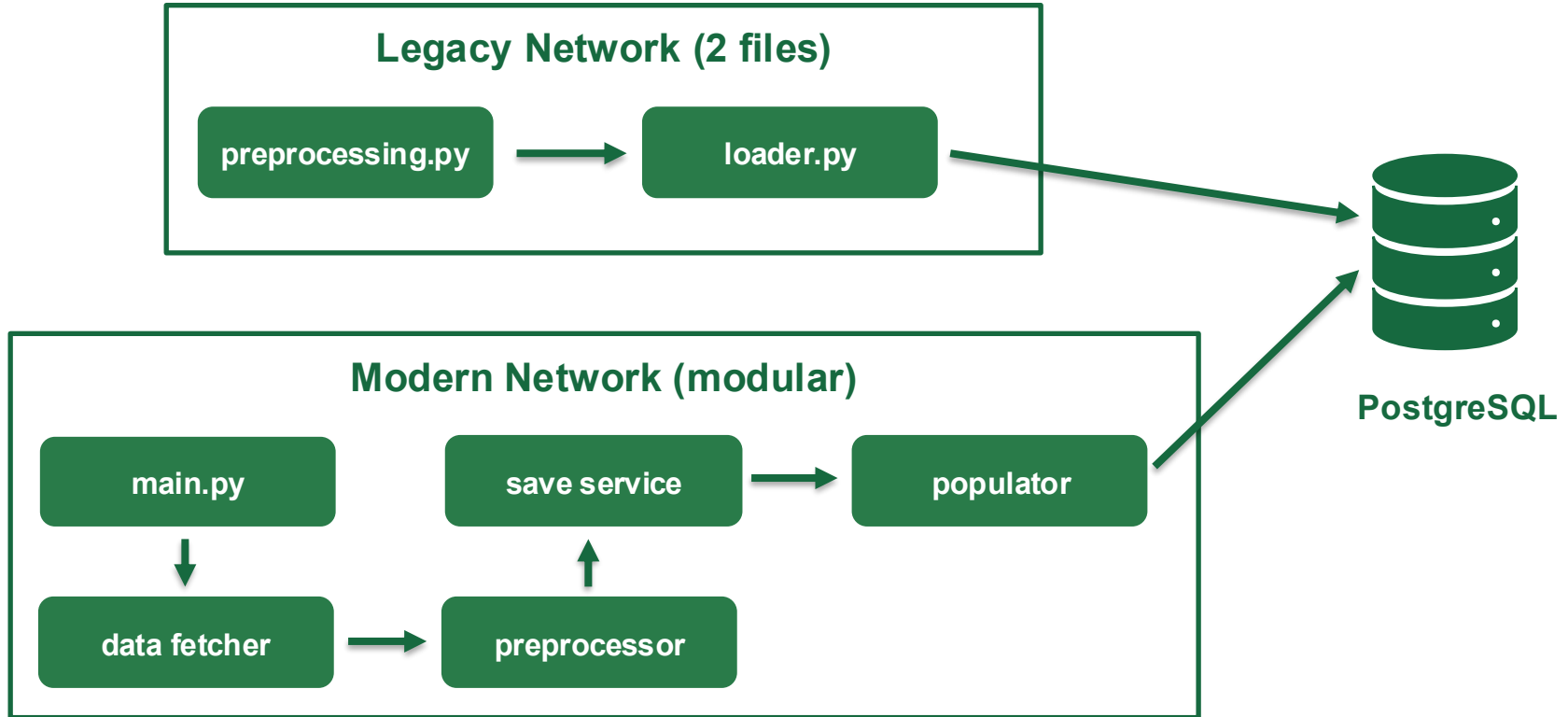
GRNsight updates data at least once per year so biologists can have the latest information

GRNsight v7.4.1

Welcome to GRNsight. Please select a network to view.

The screenshot displays the GRNsight v7.4.1 software interface. The background shows a network diagram with 15 nodes and 28 edges. The nodes are labeled: ZAP1, HMO1, SWI4, GLN3, ASH1, GCR2, SFP1, YOX1, HAP4, and CIN5. The interface includes a top menu bar with options: Network, Layout, Node, Edge, View, Export, Help, and Demo. The left sidebar contains sections for Network, Layout, and Node. The Network section shows 'Network Source' set to 'Demo' and 'Demo #1: Unweighted' selected. The Layout section shows 'Force Graph' and 'Grid Layout' buttons, a 'Link Distance (1-1000): 500' slider, a 'Charge (-2000-0): -50' slider, and checkboxes for 'Lock Force Parameters' and 'Reset Force Parameters'. The Node section shows a 'Node' label and a directional pad. A 'Load from Database' dialog box is open in the center, titled 'Generate Network'. It contains a warning: 'Warning: changing network type or source will clear the list of selected genes or proteins below.' The 'Network Type' is set to 'Gene Regulatory'. The 'Network Source' dropdown is open, showing a list of sources: 'AllianceMine - SGD: 2026-01-22' (checked), 'Yeastmine - SGD: 2024-03-19', 'Yeastmine - SGD: 2023-03-08', and 'Yeastmine - SGD: 2022-03-07'. The 'Select gene' field is empty. The 'Added genes go to' field is also empty. The dialog box has 'Generate Network' and 'Cancel' buttons at the bottom.

Database pipeline is being migrated to a modular service architecture



Outline

- GRNsight is an open-source tool that allows for easier visualization of Gene Regulatory Network (GRN) and Protein Protein Interactions (PPI).
- GRNsight has migrated from JavaScript frontend to React Vite and Grommet UI.
- We transitioned our primary data source from YeastMine to AllianceMine
- We re-engineered database pipeline into a modular architectural pattern.
- The primary goal of this modernization is to provide a faster, more scalable, and intuitive platform for biological network visualization.

React migration resulted in tech stack improvements for developer experience and shorter build time

Feature	Classic	React / New Updates
Frontend	 Bootstrap  pug  jQuery 	 grommet  React 
Backend	 node  PostgreSQL express  JS	 node  PostgreSQL express  JS
Build Tool	 webpack	 Vite
Testing	 MOCHA  chai	 Vitest
State Management	Manual DOM manipulation	Centralized React State
Performance	Slower bundle/re-renders	Instant development build and optimized production builds

Impact of database migration includes scalability and data currency



Scalability: The new modular pattern allows for easier integration of future data sources without rewriting the entire core logic.



Data Currency: Ensure biologists always have access to the latest information

Future Work

Migration	Future Work
 Load from database  Open file functionality	 Automate deployment of React app  Change server option on Amazon Web Service (AWS) to potentially less expensive one

Summary

- GRNsight is an open-source tool that allows for easier visualization of Gene Regulatory Network (GRN) and Protein Protein Interactions (PPI).
- GRNsight has migrated from JavaScript frontend to React Vite and Grommet UI.
- We transitioned our primary data source from YeastMine to AllianceMine.
- We re-engineered database pipeline into a modular architectural pattern.
- The primary result of this modernization is a faster, more scalable, and intuitive platform for biological network visualization.



GRNsight turns 12 years old today!



- Viewport and bounding box separated
- Zooming and scrolling
- Edge weight normalization factor

- Model-View-Controller paradigm introduced

- Backend GRN database from YeastMine

- Port to React
- Load database from Allancemine



- GRNsight server live on **March 20, 2014**
- Loads unweighted and weighted GRNs using D3's force graph layout
- Imports Excel, SIF, GraphML formats
- Demo files provided

- Import an entire GRNmap-compatible Excel workbook
- Node coloring
- Expression database
- Gene pages
- Grid layout

- Export to GRNmap-compatible Excel workbook
- Refactor expression database

- Displays PPIs
- Backend PPI database from YeastMine
- UI and export improvements
- GitHub actions

Acknowledgements

John David N. Dionisio

Kam D. Dahlquist

Nikki C. Chun

Amelie T. Dinh

Milka Y. Zekarias

Alex J. Miller

Jia S. Garcia

Cindy L. Tong

LMU Department of Biology

HHMI STEM Research Learning Community



<http://dondi.github.io/GRNsight/>



<https://github.com/dondi/GRNsight>

References

- AZoLifeSciences. “A Guide to Understanding Gene Expression.” *AZoLifeSciences*, 27 Jan. 2021, www.azolifesciences.com/article/A-Guide-to-Understanding-Gene-Expression.aspx.
- “PDB101: Molecule of the Month: Vitamin D Receptor.” *RCSB: PDB-101*, pdb101.rcsb.org/motm/155.
- “PDB101: Molecule of the Month: Cyclin and Cyclin-dependent Kinase.” *RCSB: PDB-101*, pdb101.rcsb.org/motm/236.